12 2026-02-11 | Week 05 | Lecture 12

12.1 Examples of using independence

This is based on chapter 2.5 in the textbook.

Recall that two events A and B are **independent** if $\mathbb{P}[A \cap B] = \mathbb{P}[A]\,\mathbb{P}[B]$. This means that knowledge of one event gives no information about the other.

Example 46. The proportions of blood types in the US are

A: 40% B: 11% AB: 4% O: 45%

(a) What is the probability that two randomly-selected people are both O.

Solution: Assuming that the two blood types are independent, we have

$$\begin{aligned}\mathbb{P}[\text{person 1 has O blood} \cap \text{person 2 has O blood}] &= \mathbb{P}[\text{person 1 has O blood}]\,\mathbb{P}[\text{person 2 has O blood}] \\ &= (.45)(.45) \\ &= 0.2025.\end{aligned}$$

(b) What is the probability that two randomly-selected people have matching blood types?

Solution: Let M be the event that the people have matching blood type, and let M_X be the event that both people have type X blood, for $X \in \{A, B, AB, O\}$.

$$\begin{aligned}\mathbb{P}[M] &= \mathbb{P}[M_A \cup M_B \cup M_{AB} \cup M_O] \\ &= \mathbb{P}[M_A] + \mathbb{P}[M_B] + \mathbb{P}[M_{AB}] + \mathbb{P}[M_O],\end{aligned}$$

where the last equality followed from the fact that the events M_A, M_B, M_{AB}, M_O are pairwise disjoint. We already computed $\mathbb{P}[M_O] = (.45)(.45) = .2025$. By similar calculations, we get

$$\mathbb{P}[M] = (.40)(.40) + (.11)(.11) + (.04)(.04) + (.45)(.45) = .3762.$$

End of Example 46. □

12.2 Dicrete Random Variables

Based on sections 3.1 and 3.2 of the textbook

We often associate numbers with outcomes of an experiment.

Definition 47 (Random variables)**.** Let S be a sample space of an experiment. A ***random variable*** is any rule that associates a number with each outcome in S. In other words, it is a function whose domain is S and whose range is $\mathbb{R}$.

The value of a random variable is thought of as varying depending on the outcome of the experiment. Random variables are usually denoted with capital letters, like X, Y, Z.

Example 48. Roll a 4-sided dice twice (this is the **experiment**). There are 16 possible **outcomes**. The **sample space** is

$$S = \{(x, y) : x, y \in \{1, 2, 3, 4\}\}.$$

Let X be the sum of the two rolls. We can represent X by the following table:

		Dice 2			
		1	2	3	4
	1	2	3	4	5
Dice 1	2	3	4	5	6
	3	4	5	6	7
	4	5	6	7	8

End of Example 48. □

Definition 49 (Discrete random variable)**.** We say that a random variable X is a ***discrete random variable*** if it can assume only a finite or countably infinite number of distinct values.

Definition 50 (Probability mass function (pmf))**.** Let X be a discrete random variable. The ***probability mass function*** (or ***pmf***) of X is the function

$$p(x) = \mathbb{P}\left[X = x\right],$$

defined for every $x \in \mathbb{R}$.

Example 51. The pmf of X in Example 49 is

$$p(2) = 1/16 \quad p(3) = 2/16, \quad p(4) = 3/16, \quad p(5) = 4/16, \quad p(6) = 3/16, \quad p(7) = 2/16, \quad p(8) = 1/16$$

and $p(x) = 0$ for all other $x \in \mathbb{R}$. It is often useful to write these as a table:

x	$\mathbb{P}[X = x]$
2	1/16
3	2/16
4	3/16
5	4/16
6	3/16
7	2/16
8	1/16

End of Example 51. □

Definition 52. For a discrete random variable X, the ***expected value*** of X, denoted $\mathbb{E}\left[X\right]$, is

$$\mathbb{E}\left[X\right] = \sum_x x\mathbb{P}\left[X = x\right]$$

where the sum ranges over all distinct values that X can take.

Remark 53. $\mathbb{E}\left[X\right]$ is the long-run average of X, if we were to repeat the experiment many times.

Example 54. Let X be the random variable in Example 49. Then

$$\begin{aligned}
\mathbb{E}\left[X\right] &= \sum_x x\mathbb{P}\left[X = x\right] \\
&= 2\left(\frac{1}{16}\right) + 3\left(\frac{2}{16}\right) + 4\left(\frac{3}{16}\right) + 5\left(\frac{4}{16}\right) + 6\left(\frac{3}{16}\right) + 7\left(\frac{2}{16}\right) + 8\left(\frac{1}{16}\right) \\
&= \frac{80}{16} \\
&= 5.
\end{aligned}$$

Therefore, the sum of two 4-sided dice rolls is, on average, 5.

End of Example 54. □

Definition 55 (Cumulative distribution function (cdf))**.** Let X be any random variable. The ***cumulative distribution function*** (or ***cdf***) of X is the function

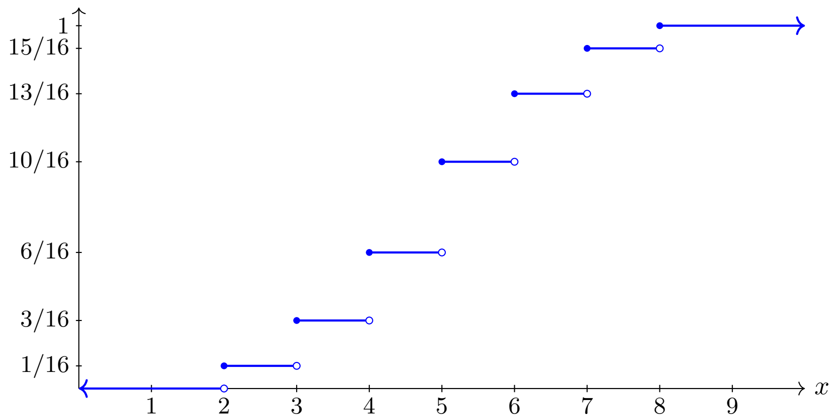
$$F(x) = \mathbb{P}\left[X \leq x\right],$$

defined for all $x \in \mathbb{R}$.

Example 56. For the random variable in Example 49, the cdf is given by the following table

x	$F(x) = \mathbb{P}[X \le x]$
2	1/16
3	3/16
4	6/16
5	10/16
6	13/16
7	15/16
8	1

Here is a plot of $F(x)$. It is the step function shown in blue:



For dicrete random variables, the graph of a cdf will always look similar to the above: it will be an increasing step function with open circles on the right endpoints and filled-in circles on the left endpoints.

End of Example 56. □